# VARUN RAMANATHAN

## Software Engineer

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Software and Machine Learning Engineer specializing in **AI-enabled Full-Stack development**, with professional experience **Designing, Developing, Testing, and Deploying** intelligent applications, **Python** automation services, and **REST APIs**. Integrated **OpenAI GPT, Claude & Google Gemini** into production financial software, implementing conversational AI, Retrieval-Augmented Generation **(RAG)**, and human-in-the-loop review and approval workflows. Experienced in building **Scalable Web Applications**, integrating APIs, developing enterprise data pipelines, and deploying solutions using **TypeScript, React, Next.js, AWS, Azure**, and **SQL**.

## EXPERIENCE

### **TD BANK** - Data Analyst (EDMO) 2025 - Present

- Engineered **Python**-based data validation and workflow automation tools using **VS Code, GitHub Copilot**, and **Agentic AI**-assisted development, reducing processing time by **over 90%** while improving data accuracy and consistency.
- Developed **SQL**-driven data quality validation pipelines to process and verify large enterprise datasets within **Microsoft Azure** cloud environments, supporting enterprise data governance and data supply chain operations.
- Designed and developed interactive **TypeScript/React Web Apps** to visualize validation results, facilitate stakeholder review, & improve workflow transparency. Built **Power BI** dashboards to monitor data quality KPIs and performance.
- Leveraged **Microsoft 365 Copilot** for workflow analysis and technical documentation, collaborating with cross-functional engineering and business teams to test, troubleshoot, and optimize automated data governance processes.

### **QUANTO** - Machine Learning Engineer 2024 - 2025

- Engineered **Python Backend** services and integrated **OpenAI GPT** and **Google Gemini APIs** into a production level AI-powered bookkeeping platform, enabling conversational financial assistance and automated financial analysis.
- Developed and optimized **LLM-powered** workflows using Retrieval-Augmented Generation **(RAG)** and model fine-tuning approaches to improve response relevance, enhance financial context retrieval and reduced Model Token usage by **over 60%**.
- Designed and implemented human-in-the-loop review and approval workflows for conversational AI agents, enabling users to validate AI-generated financial recommendations before executing bookkeeping actions.
- Developed validation and review mechanisms to improve the reliability of AI-generated financial insights and reduce the risk of incorrect automated bookkeeping actions.
- Built real-time financial data processing pipelines and developed customer-facing features using **Next.js, JavaScript, AWS, Supabase (PostgreSQL), and Vercel**, collaborating with cofounders and investors to deliver AI-powered solutions for small accounting firms.

### **CIBC** - Full-Stack Developer 2023 - 2024

- Modernized the Global Operations Workbench enterprise banking application by migrating legacy **C++** and **Java** components to **Java 21**, supporting the transition of user account management functionality to **Microsoft Azure** Cloud-based infrastructure. Designed and developed **Java Backend** services and **REST**ful **APIs** to support application integration, user account workflows, and enterprise cloud modernization initiatives.
- Developed unit and integration tests, debugged application issues, and supported User Acceptance Testing **(UAT)**, cloud deployments, and production maintenance to improve software reliability.
- Collaborated with software architects, engineering teams, and business stakeholders throughout the Software Development Life Cycle **(SDLC)**, translating requirements into scalable enterprise solutions and supporting technical design, implementation, testing, deployment, production issue resolution, and iterative enhancements.

### **McMASTER** - Machine Learning & Data Engineer 2019 - 2021

- Developed and deployed **Embedded AI** and machine learning applications using **TensorFlow, OpenCV, and STM X-CUBE-AI**, supporting research, IoT development, and industry-focused training initiatives.
- Designed and trained TensorFlow-based computer vision models for vehicle classification using aerial imagery, **achieving 95%** classification accuracy during project evaluation.
- Engineered **Python**-based automated image-labeling and intelligent traffic-analysis tools, supporting computer vision research and transportation analysis in collaboration with Peel Regional Police.
- Collaborated with academic researchers and industry partners to translate machine learning research into functional AI applications, embedded system prototypes, and technical training tools.

## TECHNICAL PROJECTS

**ADAPTIVE CHESS TRAINER** | React, TypeScript, Stockfish, Supabase, Vercel, Agentic AI
chess.varunsr.com/app • github.com/VCRAM99/adaptive-chess-trainer

- Designed, developed, and deployed a full-stack **React/TypeScript** chess training application featuring Stockfish-powered adaptive opponents, real-time player skill estimation, post-game analysis, and Supabase Authenticated custom user experiences.
- Leveraged **Cursor, Claude, and ChatGPT** in an agentic, human-supervised development workflow, reviewing AI-assissted code, implementing automated tests, and maintaining human oversight of architectural and deployment decisions.

**SIREN SENSE** | React Native, JavaScript, Expo, Machine Learning

- Developed a cross-platform mobile application using machine learning to recognize emergency sirens and hazardous sounds, with planned device-audio alerts to improve situational awareness.

## TECHNICAL SKILLS

**Languages:** Python, Java, TypeScript, JavaScript, SQL, C++, C#, HTML5, CSS3

**AI / Machine Learning:** OpenAI API, Claude API, Google Gemini API, LLM Integration, Retrieval-Augmented Generation (RAG), Agentic AI, Human-in-the-Loop Workflows, TensorFlow, OpenCV, GitHub Copilot, Cursor

**Full-Stack Development:** React, Next.js, Node.js, React Native, REST APIs, Backend Services, API Integration

**Cloud / Databases / Tools:** Microsoft Azure, AWS, Google Cloud Platform, Supabase, MySQL / PostgreSQL, Vercel, Docker, Git, PyTest, Vitest, Power BI

## EDUCATION

**McMaster University** | Bachelor of Engineering (B.Eng.) - Computer Engineering 2017–2021

**University of Michigan, Ann Arbor** | Graduate Studies in Computer Engineering 2021–2023

- Focus: Machine Learning and Data Science

## CERTIFICATIONS

**Anthropic** Claude 101 • **Anthropic** Claude Code • **Google AI** Essentials • **Github** Foundations

*More Projects showcased on **Website & Github** - **References** Available upon request*